Infrastructure guidance may smooth autonomous intersection crossings

A proof-of-concept campus testbed compared crossings with and without Mobility operating system (mOS) speed suggestions at an unsignalized intersection.

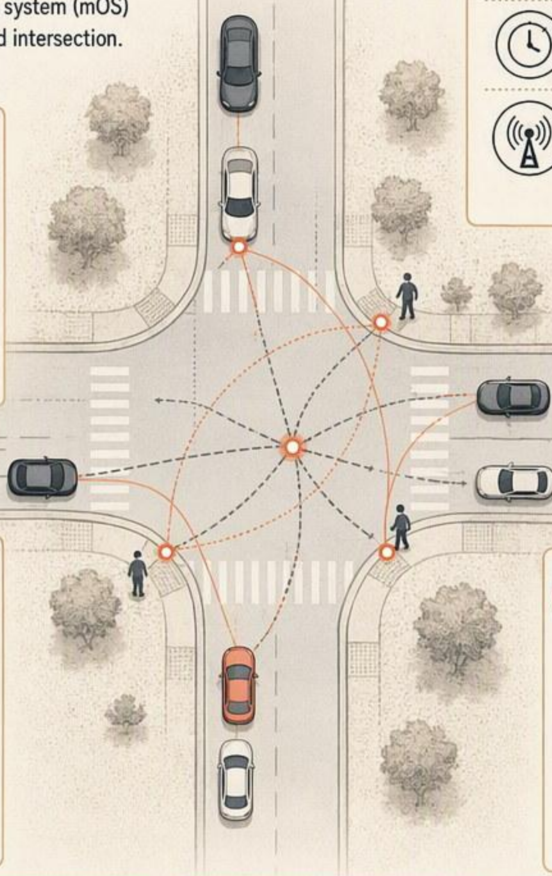
KEY STATS

4.62 s vs 5.99 s – mean Post encroachment time (PET), Vehicle-to-vehicle (V2V), with vs without speed suggestion

1.76 s vs 2.94 s – mean Post encroachment time (PET), Vehicle-to-pedestrian (V2P), with vs without speed suggestion

48.46 ms – overall mean latency

0.201% / 0.171% / 0.106% / 0.285% – mean packet loss at 20 / 30 / 40 / 50 km/h



Why this was needed:

Vehicle-to-everything (V2X) research was often limited to simulations or constrained testbeds. Society of Automotive Engineers (SAE) J2735 message formats were described as lacking the extensibility needed for emerging autonomous vehicle services.

What the system connects:

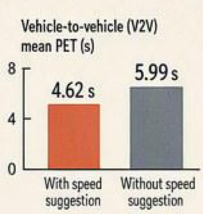
The Mobility operating system (mOS) is a modular, edge-intelligent framework. It supports plug-and-play Cellular vehicle-to-everything (C-V2X) modules, real-time intersection algorithms, and bidirectional Mixed reality (MR) interaction between physical and virtual agents.

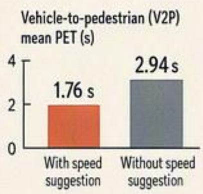
One-campus proof of concept:

The evaluation compared Vehicle-to-vehicle (V2V) and Vehicle-to-pedestrian (V2P) scenarios with and without mOS at one unsignalized four-leg intersection. The testbed was at KAIST Munji campus in Daejeon, Republic of Korea.

Smoother crossings in this test:

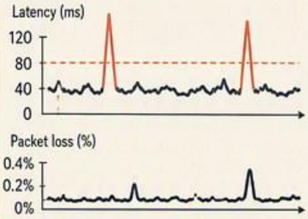
With speed suggestions, mean Post encroachment time (PET) was lower in both scenarios: 4.62 s vs 5.99 s for Vehicle-to-vehicle (V2V), and 1.76 s vs 2.94 s for Vehicle-to-pedestrian (V2P). The paper interpreted lower PETs as smoother, more coordinated interactions, while noting reported speed-suggestion PET values remained above the cited 1.0 s safety threshold.





Real-time signals, with spikes:

Average network results supported real-time operation in this setup. The authors also reported occasional spikes that could threaten stability in critical communications.



What still needs testing:

The authors conclude that mOS overcame limitations of J2735-style binary messages while maintaining real-time performance. They note that testing used a commercial Fifth-generation (5G) network rather than dedicated C-V2X infrastructure, and direct comparisons and scaling remain future work.